

Kolmogorov-Smirnov-test.R

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empirical_CDF_and_Kolmogorov_Test

Instruction: Study the empirical cumulative distribution function (ECDF) and the Kolmogorov–Smirnov test. Test whether a dataset follows a given distribution, or whether two samples come from the same one. Visualize ECDFs (implemented by yourself) and interpret.

for following the project on your own clone by:

```
git clone https://github.com/Aschir/empirical_CDF_and_Kolmogorov_Test/
```

theory about Kolmogorov-Smirnov test (KS-Test)

One-sample Kolmogorov-Smirnov Test

The one-sample Kolmogorov–Smirnov (KS) test quantifies the difference between the empirical distribution of a dataset and a theoretical distribution and can be applied when your analysed data is iid.

Example:

Testing if $X \sim N(\mu, \sigma^2)$

A basketball player Mike, shoots each training 1000 times. he has 200 trainings and scores with a probability of $p=0.5$, does his score follow a normal distribution?

$$training \sim Binom(n = 1000, p = 0.5)$$

with 100 performed trainings:

```
set.seed(42)
# data
n = 1000
prob = 0.5
n_training = 100

X = rbinom(n_training, size = n, prob = prob)
```

$$H_0 : X \sim N(\mu = 500, \sigma^2)$$

$$H_1 : X \sim N(\mu = 500, \sigma^2)$$

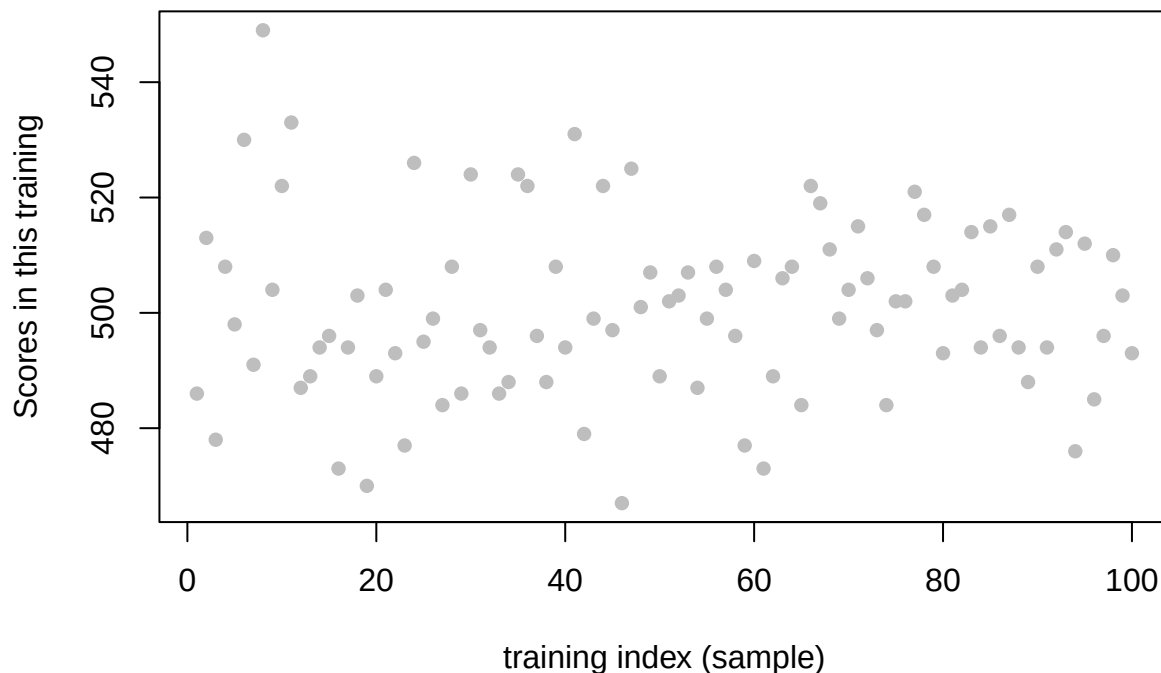
Let's have a look in it's distribution:

```
set.seed(42)

# for visualisation: 10% quantile and closest point to 10% quantile
q10 <- quantile(X, 0.1)
closest_q10 <- X[which.min(abs(X - q10))]

# ---- Plot 1: the distribution of X ----
plot(1:length(X), X,
     xlab = "training index (sample)",
     ylab = "Scores in this training",
     main = "Distribution of scores in the progress of trainings.",
     pch = 16,
     col="grey")
```

Distribution of scores in the progress of trainings.



```
# ---- Plot 2: Histogram + part shaded + theoretical curve ----
h = hist(X,
         breaks = 50,
```

```

    probability = TRUE,
    main = paste0("Distribution of scored points"),
    xlab = "Scores in a training",
    col = "lightgray",
    border = "white")

# --- Shade bars up to closest 10% quantile ---
for (i in 1:(length(h$breaks)-1)) {
  left  <- h$breaks[i]
  right <- h$breaks[i+1]
  mid   <- (left + right) / 2

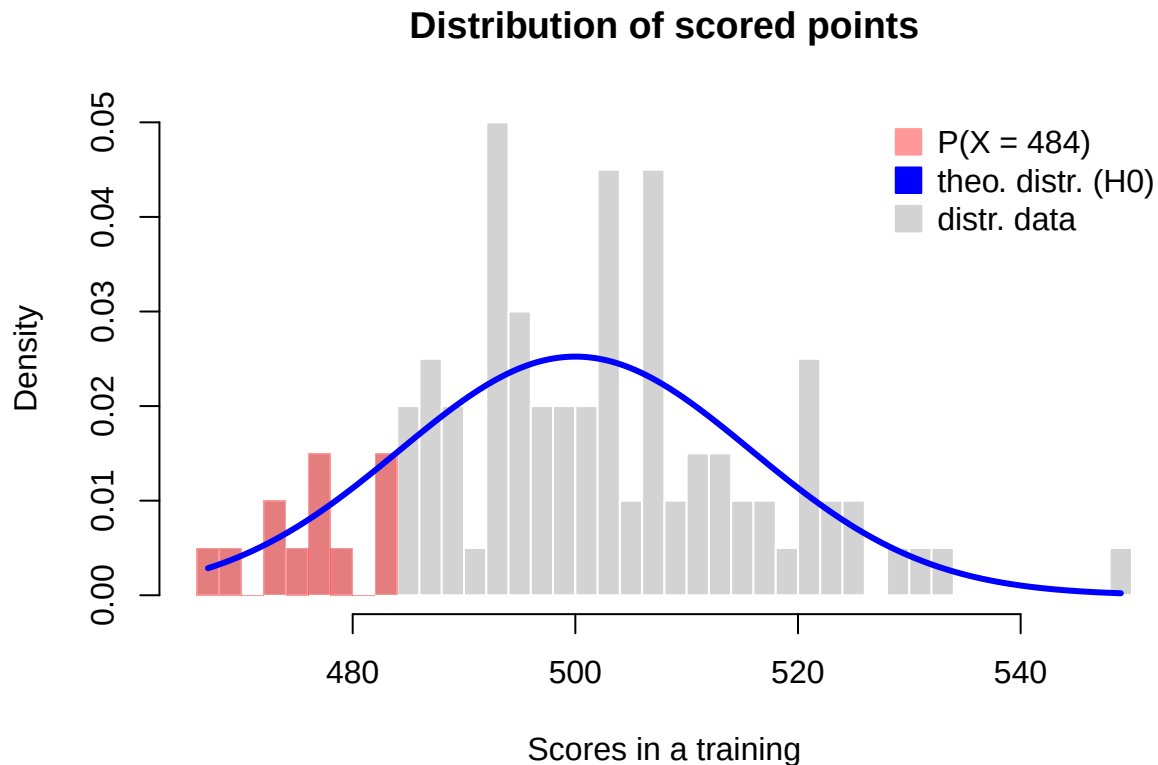
  if (mid <= closest_q10) {
    rect(left, 0, right, h$density[i],
         col = rgb(1, 0, 0, 0.4),
         border = NA)
  }
}

# --- Theoretical distribution ---
x_vals <- seq(min(X), max(X), length.out = 200)
theo_density <- dnorm(x_vals, mean = 500, sd = sqrt(250))

lines(x_vals, theo_density, col = "blue", lwd = 3)

# --- Legend ---
legend(
  "topright",
  legend = c(paste0("P(X  ", closest_q10, ")"),
             "theo. distr. (H0)",
             "distr. data"),
  pch     = 22,
  pt.cex  = 2,
  pt.bg   = c(rgb(1,0,0,0.4), "blue", "lightgray"), # fill colors
  col     = c(rgb(1,0,0,0.4), "blue", "lightgray"), # border colors
  bty     = "n"
)

```



In a general setting,

Null hypothesis: $H_0 : F_X = F_0$

Alternative hypothesis: $H_1 : F_X \neq F_0$

and our test statistic is:

$$D_n = \sup_x |F_n(x) - F(x)|$$

We make no assumptions on the parameter of the given distribution! Only the CDF

We Compute the CDF:

```
set.seed(42)

#CDF (ECDF) of the given DATA.
ecdf_X = ecdf(X) # we use the given func for the data

plot(ecdf_X,
     main = "Empirical CDF of X",
     xlab = "x",
     ylab = "F(x)",
     verticals = TRUE,
     do.points = FALSE)

# theoretical cdf in manual computation of the cdf
x_vals = seq(min(X), max(X), length.out = 200)
```

```

cdf_theo <- pnorm(x_vals, mean = 500, sd = sqrt(250))

lines(x_vals, cdf_theo, col = "blue", lwd = 2)

# Add the point seen in the previous data
y_q10 <- ecdf_X(closest_q10)

points(closest_q10, y_q10, col = rgb(1, 0, 0, 0.4), pch = 16)

# ---- Add vertical line for sup difference (Kolmogorov D) ----

# max difference computation
emp_vals <- ecdf_X(x_vals)
diffs <- abs(emp_vals - cdf_theo)
D <- max(diffs)

x_value_f <- x_vals[which.max(diffs)]
y_emp_at_f <- emp_vals[which.max(diffs)]
y_theo_at_f <- cdf_theo[which.max(diffs)]
eps <- 1e-8

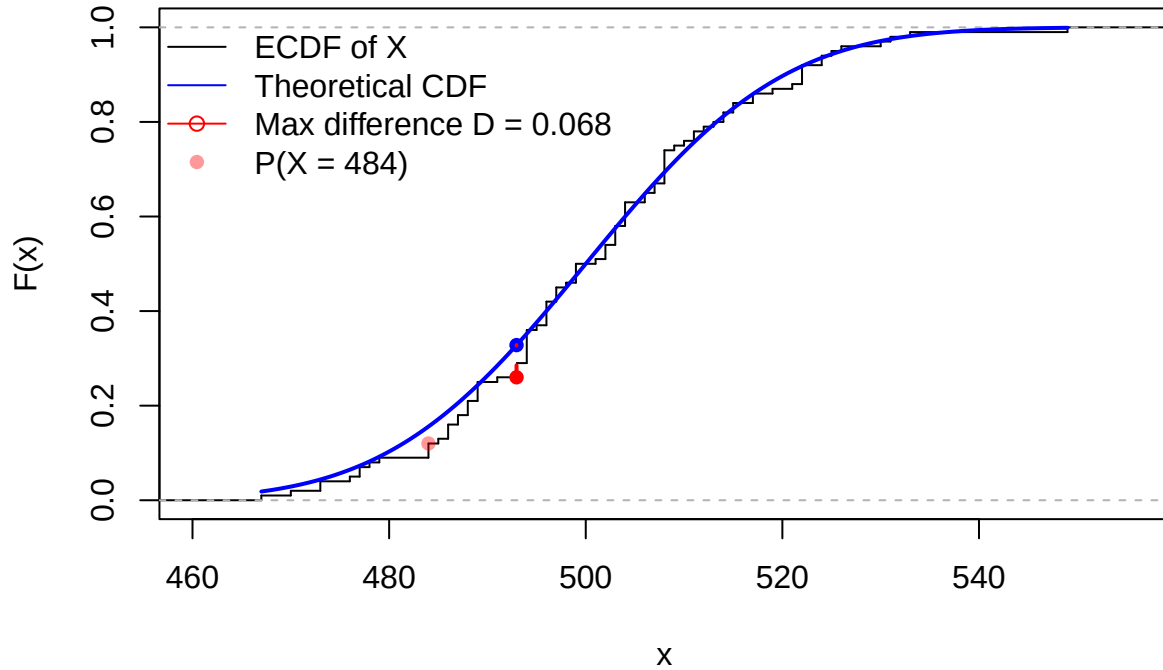
# draw the vertical line of max difference
#segments(x0 = x_value_f, y0 = y_emp_at_f,
#         x1 = x_value_f, y1 = y_theo_at_f,
#         col = "red", lwd = 2, lty = 2)

# mark max difference points in their distributions
points(x_value_f, y_emp_at_f, col="red", pch=16)
points(x_value_f, y_theo_at_f, col="blue", pch=16)
lines(x = c(x_value_f-eps, x_value_f+eps), y = c(y_emp_at_f, y_theo_at_f),
      col = "red", lwd = 2, lty = 2)

legend("topleft",
      legend = c("ECDF of X",
                  "Theoretical CDF",
                  paste0("Max difference D = ", round(D, 3)),
                  paste0("P(X ", closest_q10, ")")),
      col = c("black", "blue", "red", rgb(1,0,0,0.4)),
      lwd = c(1, 1, 1, NA),
      lty = c(1, 1, 1, NA),
      pch = c(NA, NA, 1, 16),
      bty = "n")

```

Empirical CDF of X



we used the maximum as a numerical approximation of the supremum.

the supremum would consider all points not only the ones in the dataset.

the maximum deviation difference is not normal distributed but follows a Kolmogorov-Smirnov deviations-distribution defined by:

The maximum deviation D_n is **not normally distributed**. Instead, it follows the **Kolmogorov distribution**, which has the cumulative distribution function:

$$P(\sqrt{n}D_n \leq \lambda) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 \lambda^2}$$

which is in r:

```
ks_pvalue <- function(D, n, maxK=1000){
  lambda <- sqrt(n) * D
  k <- 1:maxK
  pval <- 2 * sum((-1)^(k-1) * exp(-2 * k^2 * lambda^2))
  pval <- min(max(pval, 0), 1)
  return(pval)
}
```

now we compute the p-value: and form our decision boundary:

$$P(D_n \geq D_{\text{obs}} \mid H_0)$$

```
p_value <- ks_pvalue(D, length(X))
p_value
```

```
## [1] 0.7431522
```

```
# Decision based on alpha
alpha <- 0.05
if(p_value < alpha){
  cat("Reject H0: the data does not follow the theoretical distribution.\n")
} else {
  cat("Fail to reject H0: no evidence against the theoretical distribution.\n")
}
```

```
## Fail to reject H0: no evidence against the theoretical distribution.
```

we can compare our computation with the builtin-method:

the results are not very accurate since we used the maximum instead of the minimum and for sakes of explainability an discrete distribution function instead of an continious function

```
# Built-in KS test
ks_result <- ks.test(X, "pnorm", mean = 500, sd = sqrt(250))
```

```
## Warning in ks.test.default(X, "pnorm", mean = 500, sd = sqrt(250)): ties should
## not be present for the one-sample Kolmogorov-Smirnov test
```

```
ks_result
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: X
## D = 0.068985, p-value = 0.7281
## alternative hypothesis: two-sided
```

Project Continuity: Theory vs. Application (The K-S Test)

Parameter	Theoretical Component	Practical Component
Data Source	Used an artificially generated dataset (Binomial simulation) to ensure clean, controlled data for formula testing.	Will use a real-world dataset (faithful geyser waiting times) known to be messy and bimodal, representing a true challenge.
Primary Goal	Confirmed a hypothesis (Failed to reject the Null Hypothesis), proving the K-S test correctly identifies a clean, theoretical distribution.	Will challenge a hypothesis (Will reject the Null Hypothesis), proving that the real data is too complex for simple models and requires deeper analysis.
Core Method	Focused on formula validation by manually calculating the K-S P-value	Will include the manual ECDF Plot and the Two distinct distribution plot
ECDF Visualization	Visualized the ECDF of the observed data against the theoretical distribution to check for fit	Will manually create the ECDF step chart to prove foundational understanding of how the data accumulates.

Also, we will run both the One-Sample and the Two-Sample K-S Tests on the practical geyser data to prove it is not a single bell curve, but is actually made of two separate and distinct schedules

Project Goals For this messy, bimodal, real world dataset

- **Verifying Distribution:** We use statistics to formally prove that the geyser data, which visibly shows two different schedules (short and long waits), is **not** a simple, single "bell curve."
- **Manual ECDF Implementation:** We manually build the cumulative step chart (ECDF) to show exactly how the data accumulates from 0% to 100%.
- **One-Sample K-S Test:** We run this test to **reject** the bell curve assumption, stopping us from making the critical mistake of using a simple, wrong forecasting model.
- **Two-Sample K-S Test:** We split the data into two groups (Short Waits and Long Waits) and use this test to **prove** that they are drawn from two separate and distinct distributions.

Advantages and Applications (Why It Matters)

- **Prevents Model Failure:** The most critical advantage is preventing the misapplication of statistical models. Ignoring the bimodal nature of the data would lead to unreliable forecasting results if a simple Normal model were used

STEPS

Project 4: Empirical CDF and Kolmogorov Tests - Steps

1. **Set Up the Environment:** Started coding on blank R Script in RStudio
2. **Load Data and Library:** In this step, I loaded the **ggplot2** plotting tool and the **faithful** dataset. After quickly inspecting the data structure (`str`, `head`), I extracted the vital waiting column and saved it specifically as **waiting_times**. The step concluded by running a summary to establish the initial statistics for this new variable.

`str(faithful)` : This shows you the names of the variables and their types

```
> data(faithful)
> str(faithful)
'data.frame':  272 obs. of  2 variables:
 $ eruptions: num  3.6 1.8 3.33 2.28 4.53 ...
 $ waiting  : num  79 54 74 62 85 55 88 85 51 85 ...
> |
```

`Head(faithful)` : View the first few rows to understand the data

```
> head(faithful)
  eruptions waiting
1    3.600      79
2    1.800      54
3    3.333      74
4    2.283      62
5    4.533      85
6    2.883      55
> |
```

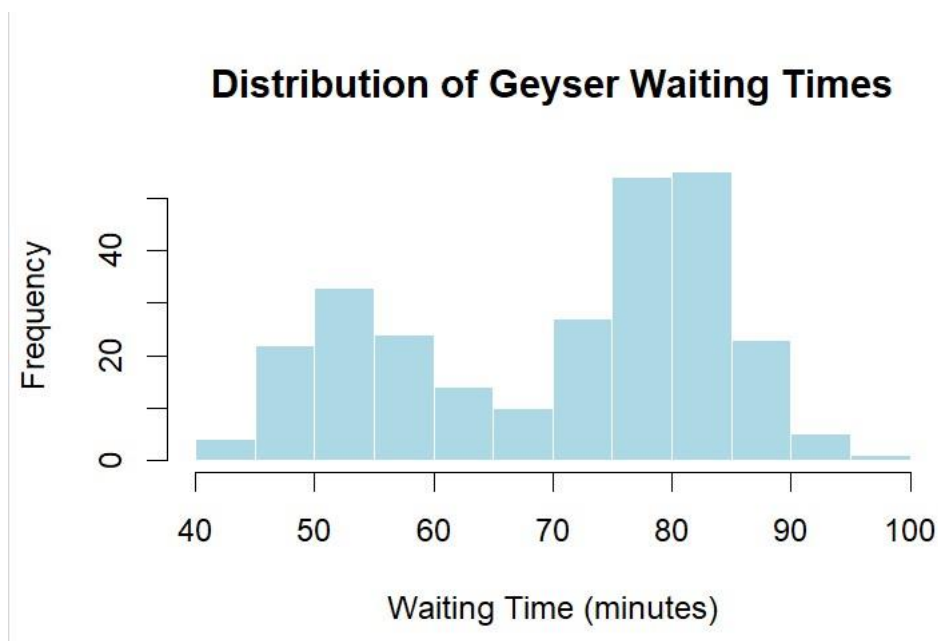
Basically , Inspecting the structure of the dataset

Then, got descriptive statistics of the variable 'waiting_times'

```
waiting_times <- faithful$waiting
# 5. Get descriptive statistics
summary(waiting_times)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  43.0   58.0   76.0   70.9   82.0   96.0
|
```

3. **Inspect Data:** Plotted a histogram on waiting_times to visually confirm its bimodal nature, which indicates it is not a Normal distribution.

```
hist(waiting_times,
     main = "Distribution of Geyser Waiting Times",
     xlab = "Waiting Time (minutes)",
     col = "lightblue",
     border = "white")
|
```



And this makes our project interesting , Why ?

Why This Makes the Project "Interesting":

- Because the histogram shows two humps, it tells us that the geyser is not operating on one single, random schedule. It strongly suggests there are **two different natural states**: a "short cycle" (around 55 minutes) and a "long cycle" (around 80 minutes).

- Our project's entire purpose is to take that **visual suggestion** from the histogram and use advanced statistical tools (the ECDF and K-S tests) to **mathematically prove** that those two schedules are real and separate.

4. **Sort the Data (ECDF Step 1):** Sort the data points, which will serve as the x-values for the ECDF plot.

```
data_sorted <- sort(waiting_times)
```

data_sorted	num [1:272]	43	45	45	45	46	46	46	46	46	4...
waiting_times	num [1:272]	79	54	74	62	85	55	88	85	51	8...

5. **Calculate Proportions (ECDF Step 2):** Determine the total number of observations (n) and calculate the cumulative proportion (i/n) for each sorted point.

```
n <- length(data_sorted)
```

```
ecdf_values <- (1:n) / n
```

Values											
data_sorted	num [1:272]	43	45	45	45	46	46	46	46	46	4...
ecdf_values	num [1:272]	0.00368	0.00735	0.01103	0.01471	0.01839	0.02207	0.02574	0.02942	0.03310	0.03677
n	272L										
waiting_times	num [1:272]	79	54	74	62	85	55	88	85	51	8...

6. **Prepare Plotting Data:** Combined the sorted data and the proportions into a single data frame for visualization.

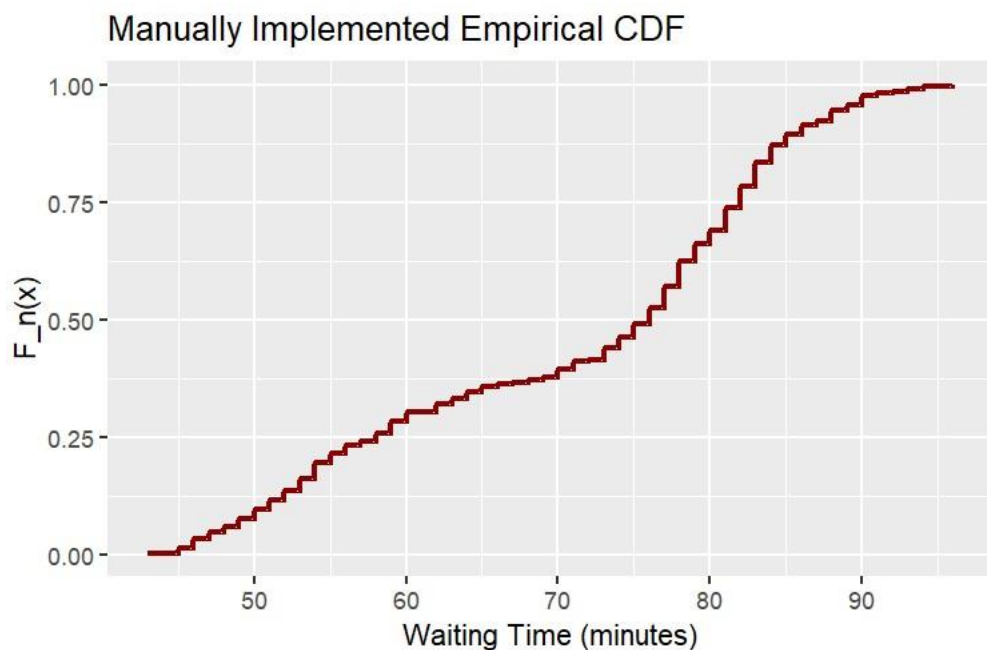
```
ecdf_df_manual <- data.frame(
```

```
  x = data_sorted,
```

```
  Fn = ecdf_values
```

```
)
```

7. **Visualize the ECDF:** Created a plot using ggplot2 and geom_step() to generate the required step-function visualization of the manually calculated ECDF.



The chart is mathematically perfect because it correctly plots the cumulative percentage from 0% to 100% using steps, and clearly shows the two distinct clusters where the curve rises steeply.

8. **Estimate Normal Parameters:** Calculate the mean (μ) and standard deviation (σ) of the data, as these are required parameters for the theoretical Normal distribution in the K-S test.

```
mu <- mean(waiting_times)
```

```
sigma <- sd(waiting_times)
```

μ	70.8970588235294
n	272L
σ	13.5949737899994

9. **Performed One-Sample K-S Test:** Executed the test against the theoretical Normal CDF (pnorm) defined by the calculated μ and σ .

```
ks_test_normal <- ks.test(waiting_times, "pnorm", mean = mu, sd = sigma)
```

```
print(ks_test_normal)
```

```
> print(ks_test_normal)
```

```
Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data: waiting_times
```

```
D = 0.15536, p-value = 3.967e-06
```

```
alternative hypothesis: two-sided
```

10. **Interpret K-S Test 1:** #The results from the one-sample Kolmogorov-Smirnov test show a test statistic of **$D = 0.15536$** and a **$p\text{-value} = 3.967e-06$** .

#Since the p-value (0.000003967) is far below the conventional significance level of $\alpha = 0.05$, we must **reject the Null Hypothesis (H_0)** that the data is drawn from a Normal distribution defined by the sample's mean and standard deviation.

#This statistical rejection strongly validates our initial visual assessment of the histogram, which showed the data is clearly **bimodal** (has two peaks), making it fundamentally incompatible with the single-peaked, bell-shaped characteristics of the Normal distribution.

11. **Create Two Samples:** Split the data into two distinct groups based on the median to prepare for the two-sample K-S test.

- `sample_A <- waiting_times[waiting_times <= median_time]`
- `sample_B <- waiting_times[waiting_times > median_time]`

sample_A	num [1:143]	54 74 62 55 51 54 47 52 62 5...
sample_B	num [1:129]	79 85 88 85 85 84 78 83 84 7...

```
> print(paste("Median Split Point:", median_time, "minutes"))
[1] "Median Split Point: 76 minutes"
> print(paste("Sample A (Short) Size:", length(sample_A)))
[1] "Sample A (Short) Size: 143"
> print(paste("Sample B (Long) Size:", length(sample_B)))
[1] "Sample B (Long) Size: 129"
```

12. **Performed Two-Sample K-S Test:** Run the K-S test to check if sample_A and sample_B are drawn from the same continuous distribution.

```
Asymptotic two-sample Kolmogorov-Smirnov test

data: sample_A and sample_B
D = 1, p-value < 2.2e-16
alternative hypothesis: two-sided
```

13. **Interpret K-S Test 2:** #The Two-Sample Kolmogorov-Smirnov test was performed to check the Null Hypothesis (H_0) that if "shorter waiting times" (Sample A) and "longer waiting times" (Sample B) come from the same continuous distribution.

#The result is highly significant:

#* K-S Statistic (D): 1

#P-value: < 2.2e-16

#The test statistic D achieved its maximum value of 1, and the resulting p-value is effectively zero, we strongly reject the Null Hypothesis.

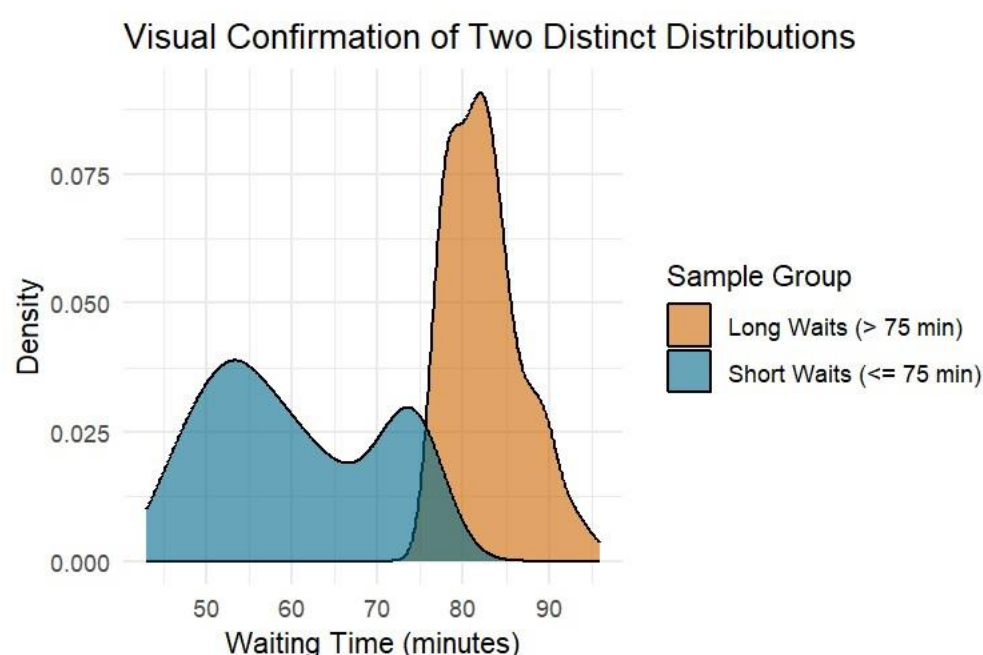
#Conclusion: The distribution of the short waiting times and the distribution of the long waiting times are statistically distinct. This confirms that the `faithful` dataset is not just bimodal, but is likely a mixture of two separate (two different waiting time distributions operating within the same system).

14. Visualize the Two Separate Distributions (for pedagogical completeness)

```
# 1. Create a data frame for Sample A and add a group label
df_A <- data.frame(
  waiting = sample_A,
  group = "Short Waits (<= 75 min)"
)

# 2. Create a data frame for Sample B and add a group label
df_B <- data.frame(
  waiting = sample_B,
  group = "Long Waits (> 75 min)"
)

# 3. Combine the two data frames
combined_samples_df <- rbind(df_A, df_B)
```



#Why this is a Great Result ?

#The graph clearly shows why we rejected the Null Hypothesis in both K-S tests

#We rejected the idea that the total data follows one Normal curve.

#The image shows the two separate bell shapes that are actually present.

#Two-Sample K-S Test (Step 12): You rejected the idea that the "Short Waits" (Blue) and "Long Waits" (Orange) come from the same distribution.

#The graph clearly shows these two distributions are distinct and separate.

#Confirms $D=1$: The minimal overlap between the blue and orange curves, especially in the tails, is what drove your K-S statistic D to 1.